

- 3.95 Plan: Remember that molarity is moles of solute in volume of solution.  
Solution: Volumes may **not** be additive when two different solutions are mixed, so the final volume may be slightly different from 1000.0 mL. The correct method would state, "Take 100.0 mL of the 10.0 M solution and add water until the total volume is 1000 mL."

- 3.96 Plan: In all cases, the definition of molarity ( $M = \frac{\text{moles solute}}{\text{L of solution}}$ ) will be important. Volume must be expressed in liters. The molar mass is used to convert moles to grams. The chemical formulas must be written to determine the molar mass.

Solution:

$$\text{a) Grams Ca(C}_2\text{H}_3\text{O}_2)_2 = (185.8 \text{ mL}) \left( \frac{10^{-3} \text{ L}}{1 \text{ mL}} \right) \left( \frac{0.267 \text{ mol Ca(C}_2\text{H}_3\text{O}_2)_2}{1 \text{ L}} \right) \left( \frac{158.17 \text{ g Ca(C}_2\text{H}_3\text{O}_2)_2}{1 \text{ mol Ca(C}_2\text{H}_3\text{O}_2)_2} \right)$$

$$= 7.84659 = \mathbf{7.85 \text{ g Ca(C}_2\text{H}_3\text{O}_2)_2}$$

$$\text{b) Moles KI} = (21.1 \text{ g KI}) \left( \frac{1 \text{ mol KI}}{166.0 \text{ g KI}} \right) = 0.127108 \text{ moles KI}$$

$$\text{Volume} = (500. \text{ mL}) \left( \frac{10^{-3} \text{ L}}{1 \text{ mL}} \right) = 0.500 \text{ L}$$

$$\text{Molarity KI} = \frac{0.127108 \text{ mol KI}}{0.500 \text{ L}} = 0.254216 = \mathbf{0.254 \text{ M KI}}$$

$$\text{c) Moles NaCN} = \left( \frac{0.850 \text{ mol NaCN}}{1 \text{ L}} \right) (145.6 \text{ L}) = 123.76 = \mathbf{124 \text{ mol NaCN}}$$

- 3.100 Plan: These are dilution problems. Dilution problems can be solved by converting to moles and using the new volume, however, it is much easier to use  $M_1V_1 = M_2V_2$ . Part (c) may be done as two dilution problems or as a mole problem. The dilution equation does not require a volume in liters; it only requires that the volume units match.

Solution:

$$\text{a) } M_1 = 0.250 \text{ M KCl} \quad V_1 = 37.00 \text{ mL} \quad M_2 = ? \quad V_2 = 150.00 \text{ mL}$$

$$M_1V_1 = M_2V_2 \quad (0.250 \text{ M})(37.00 \text{ mL}) = (M_2)(150.0 \text{ mL})$$

$$\frac{(0.250 \text{ M})(37.00 \text{ mL})}{150.0 \text{ mL}} = M_2 \quad M_2 = 0.061667 = \mathbf{0.0617 \text{ M KCl}}$$

$$\text{b) } M_1 = 0.0706 \text{ M (NH}_4)_2\text{SO}_4 \quad V_1 = 25.71 \text{ mL} \quad M_2 = ? \quad V_2 = 500.00 \text{ mL}$$

$$M_1V_1 = M_2V_2 \quad (0.0706 \text{ M})(25.71 \text{ mL}) = (M_2)(500.0 \text{ mL})$$

$$\frac{(0.0706 \text{ M})(25.71 \text{ mL})}{500.0 \text{ mL}} = M_2 \quad M_2 = 0.003630 = \mathbf{0.00363 \text{ M (NH}_4)_2\text{SO}_4}$$

c) When working this as a mole problem it is necessary to find the individual number of moles of sodium ions in each separate solution. (Rounding to the proper number of significant figures will only be done for the final answer.)

$$\text{Moles Na}^+ \text{ from NaCl solution} = (3.58 \text{ mL}) \left( \frac{10^{-3} \text{ L}}{1 \text{ mL}} \right) \left( \frac{0.348 \text{ mol NaCl}}{1 \text{ L}} \right) \left( \frac{1 \text{ mol Na}^+}{1 \text{ mol NaCl}} \right)$$

$$= 0.00124584 \text{ mol Na}^+$$

$$\text{Moles Na}^+ \text{ from Na}_2\text{SO}_4 \text{ solution} = (500. \text{ mL}) \left( \frac{10^{-3} \text{ L}}{1 \text{ mL}} \right) \left( \frac{6.81 \times 10^{-2} \text{ mol Na}_2\text{SO}_4}{1 \text{ L}} \right) \left( \frac{2 \text{ mol Na}^+}{1 \text{ mol Na}_2\text{SO}_4} \right)$$

$$= 0.0681 \text{ mol Na}^+$$

$$\text{Molarity of Na}^+ = \left( \frac{(0.00124584 + 0.0681) \text{ mol Na}^+}{(3.58 + 500.) \text{ mL}} \right) \left( \frac{1 \text{ mL}}{10^{-3} \text{ L}} \right) = 0.1377058 = \mathbf{0.138 \text{ M Na}^+ \text{ ions}}$$

- 3.102 Plan: Use the density of the solution to find the mass of 1 L of solution. The 70.0% by mass translates to 70.0 g solute/100 g solution and is used to find the mass of  $\text{HNO}_3$  in 1 L of solution. Convert mass of  $\text{HNO}_3$  to moles to obtain moles/L, molarity.

Solution:

$$\text{a) Mass HNO}_3 \text{ per liter} = \left( \frac{1.41 \text{ g Solution}}{1 \text{ mL}} \right) \left( \frac{1 \text{ mL}}{10^{-3} \text{ L}} \right) \left( \frac{70.0 \text{ g HNO}_3}{100 \text{ g Solution}} \right) = 987 \text{ g HNO}_3 / \text{L}$$

$$\begin{aligned} \text{b) Molarity of HNO}_3 &= \left( \frac{1.41 \text{ g Solution}}{1 \text{ mL}} \right) \left( \frac{1 \text{ mL}}{10^{-3} \text{ L}} \right) \left( \frac{70.0 \text{ g HNO}_3}{100 \text{ g Solution}} \right) \left( \frac{1 \text{ mol HNO}_3}{63.02 \text{ g HNO}_3} \right) \\ &= 15.6617 = \mathbf{15.7 \text{ M HNO}_3} \end{aligned}$$

- 3.114 Plan: The moles of narceine and the moles of water are required. We can assume any mass of narceine hydrate (we will use 100 g), and use this mass to determine the mass of water present and convert the mass to moles of the hydrate. The mass of water will be converted to moles. Finally, the ratio of the moles of hydrate to moles of water will give the amount of water present.

Solution:

$$\begin{aligned} \text{Moles narceine hydrate} &= (100 \text{ g narceine hydrate}) \left( \frac{1 \text{ mol narceine hydrate}}{499.52 \text{ g narceine hydrate}} \right) \\ &= 0.20019 \text{ mol narceine hydrate} \end{aligned}$$

$$\begin{aligned} \text{Moles H}_2\text{O} &= (100 \text{ g narceine hydrate}) \left( \frac{10.8\% \text{ H}_2\text{O}}{100\% \text{ narceine hydrate}} \right) \left( \frac{1 \text{ mol H}_2\text{O}}{18.02 \text{ g H}_2\text{O}} \right) \\ &= 0.59933 \text{ mol H}_2\text{O} \end{aligned}$$

The ratio of water to hydrate is:  $(0.59933 \text{ mol}) / (0.20019 \text{ mol}) = 3$

Thus, there are three water molecules per mole of hydrate. The formula for narceine hydrate is **narceine $\cdot$ 3H<sub>2</sub>O**.

- 3.134 Plan: This problem may be done as two dilution problems with the two final molarities added, or, as done here, it may be done by calculating, then adding the moles and dividing by the total volume.

Solution:

$$\begin{aligned} M \text{ KBr} &= \frac{\text{Total Moles KBr}}{\text{Total Volume}} = \frac{\text{Moles KBr from Solution 1} + \text{Moles KBr from Solution 2}}{\text{Volume Solution 1} + \text{Volume Solution 2}} \\ M \text{ KBr} &= \frac{\left( \frac{0.053 \text{ mol KBr}}{1 \text{ L}} \right) (0.200 \text{ L}) + \left( \frac{0.078 \text{ mol KBr}}{1 \text{ L}} \right) (0.550 \text{ L})}{0.200 \text{ L} + 0.550 \text{ L}} = 0.071333 = \mathbf{0.071 \text{ M KBr}} \end{aligned}$$

- 3.161 Plan: Careful tracking of the units will carry you through this problem.

Solution:

$$\text{a) Dissolved salt} = (50.0 \text{ mL H}_2\text{O}) \left( \frac{10^{-3} \text{ L H}_2\text{O}}{1 \text{ mL H}_2\text{O}} \right) \left( \frac{2.50 \text{ kg Salt}}{1 \text{ L H}_2\text{O}} \right) \left( \frac{10^3 \text{ g}}{1 \text{ kg}} \right) = \mathbf{125 \text{ g salt}}$$

b) Convert the grams from part (a) to moles of salt, then to moles of cocaine, and finally to grams of cocaine. A calculation similar to that in part (a) will give the volume required.

$$\begin{aligned} \text{Mass cocaine} &= (125 \text{ g Salt}) \left( \frac{1 \text{ mol Salt}}{339.81 \text{ g Salt}} \right) \left( \frac{1 \text{ mol Cocaine}}{1 \text{ mol Salt}} \right) \left( \frac{303.35 \text{ g Cocaine}}{1 \text{ mol Cocaine}} \right) \\ &= 111.588 \text{ g cocaine (unrounded)} \end{aligned}$$

$$\begin{aligned} \text{Additional water needed} &= (111.588 \text{ g Cocaine}) \left( \frac{1 \text{ L}}{1.70 \text{ g Cocaine}} \right) - (50.0 \text{ mL}) \left( \frac{10^{-3} \text{ L}}{1 \text{ mL}} \right) \\ &= 65.59 = \mathbf{65.6 \text{ L H}_2\text{O}} \end{aligned}$$

- 4.2 Ionic and polar covalent compounds are most likely to be soluble in water. Because water is polar, the partial charges in water molecules are able to interact with the charges, either ionic or dipole-induced, in other substances.

- 4.3 Ions must be present in an aqueous solution for it to conduct an electric current. Ions come from ionic compounds or from other electrolytes such as acids and bases.

- 4.10 a) Benzene is likely to be insoluble in water because it is non-polar and water is polar.  
 b) Sodium hydroxide, an ionic compound, is likely to be soluble in water since the ions from sodium hydroxide will be held in solution through ion-dipole attractions with water.  
 c) Ethanol ( $\text{CH}_3\text{CH}_2\text{OH}$ ) will likely be soluble in water because the alcohol group ( $-\text{OH}$ ) will hydrogen bond with the water.  
 d) Potassium acetate, an ionic compound, will likely dissolve in water to form sodium ions and acetate ions that are held in solution through ion-dipole attractions to water.

- 4.12 a) An aqueous solution that contains ions conducts electricity.  $\text{CsI}$  is a soluble ionic compound, and a solution of this salt in water contains  $\text{Cs}^+$  and  $\text{I}^-$  ions. Its solution conducts electricity.  
 b)  $\text{HBr}$  is a strong acid that dissociates completely in water. Its aqueous solution contains  $\text{H}^+$  and  $\text{Br}^-$  ions, so it conducts electricity.

- 4.18 a)  $\text{AlCl}_3(s) \rightarrow \text{Al}^{3+}(aq) + 3 \text{Cl}^-(aq)$

$$\text{Moles Al}^{3+} = (100.\text{mL}) \left( \frac{10^{-3}\text{L}}{1\text{ mL}} \right) \left( \frac{2.45\text{ mol AlCl}_3}{\text{L}} \right) \left( \frac{1\text{ mol Al}^{3+}}{1\text{ mol AlCl}_3} \right) = \mathbf{0.245\text{ mol Al}^{3+}}$$

$$\text{Moles Cl}^- = (100.\text{mL}) \left( \frac{10^{-3}\text{L}}{1\text{ mL}} \right) \left( \frac{2.45\text{ mol AlCl}_3}{\text{L}} \right) \left( \frac{3\text{ mol Cl}^-}{1\text{ mol AlCl}_3} \right) = \mathbf{0.735\text{ mol Cl}^-}$$

$$\text{Al}^{3+}\text{ ions} = (0.245\text{ mol Al}^{3+}) \left( \frac{6.022 \times 10^{23}\text{ Al}^{3+}}{1\text{ mol Al}^{3+}} \right) = 1.47539 \times 10^{23} = \mathbf{1.48 \times 10^{23}\text{ Al}^{3+}\text{ ions}}$$

$$\text{Cl}^-\text{ ions} = (0.735\text{ mol Cl}^-) \left( \frac{6.022 \times 10^{23}\text{ Cl}^-}{1\text{ mol Cl}^-} \right) = 4.42617 \times 10^{23} = \mathbf{4.43 \times 10^{23}\text{ Cl}^-\text{ ions}}$$

- b)  $\text{Li}_2\text{SO}_4(s) \rightarrow 2 \text{Li}^+(aq) + \text{SO}_4^{2-}(aq)$

$$\text{Moles Li}^+ = (1.80\text{ L}) \left( \frac{2.59\text{ g Li}_2\text{SO}_4}{1\text{ L}} \right) \left( \frac{1\text{ mol Li}_2\text{SO}_4}{109.95\text{ g Li}_2\text{SO}_4} \right) \left( \frac{2\text{ mol Li}^+}{1\text{ mol Li}_2\text{SO}_4} \right) = 0.08480 = \mathbf{0.0848\text{ mol Li}^+}$$

$$\text{Moles SO}_4^{2-} = (1.80\text{ L}) \left( \frac{2.59\text{ g Li}_2\text{SO}_4}{1\text{ L}} \right) \left( \frac{1\text{ mol Li}_2\text{SO}_4}{109.95\text{ g Li}_2\text{SO}_4} \right) \left( \frac{1\text{ mol SO}_4^{2-}}{1\text{ mol Li}_2\text{SO}_4} \right) = 0.04240 = \mathbf{0.0424\text{ mol SO}_4^{2-}}$$

$$\text{Li}^+\text{ ions} = (0.084802\text{ mol Li}^+) \left( \frac{6.022 \times 10^{23}\text{ Li}^+}{1\text{ mol Li}^+} \right) = 5.106787 \times 10^{22} = \mathbf{5.11 \times 10^{22}\text{ Li}^+\text{ ions}}$$

$$\text{SO}_4^{2-}\text{ ions} = (0.042401\text{ mol SO}_4^{2-}) \left( \frac{6.022 \times 10^{23}\text{ SO}_4^{2-}}{1\text{ mol SO}_4^{2-}} \right) = 2.55339 \times 10^{22} = \mathbf{2.55 \times 10^{22}\text{ SO}_4^{2-}\text{ ions}}$$

- c)  $\text{KBr}(s) \rightarrow \text{K}^+(aq) + \text{Br}^-(aq)$

$$\text{K}^+\text{ ions} = (225\text{ mL}) \left( \frac{10^{-3}\text{L}}{1\text{ mL}} \right) \left( \frac{1.68 \times 10^{22}\text{ F.U.KBr}}{\text{L}} \right) \left( \frac{1\text{ K}^+}{1\text{ F.U.KBr}} \right) = \mathbf{3.78 \times 10^{21}\text{ K}^+\text{ ions}}$$

$$\text{Br}^-\text{ ions} = (225\text{ mL}) \left( \frac{10^{-3}\text{L}}{1\text{ mL}} \right) \left( \frac{1.68 \times 10^{22}\text{ F.U.KBr}}{\text{L}} \right) \left( \frac{1\text{ Br}^-}{1\text{ F.U.KBr}} \right) = \mathbf{3.78 \times 10^{21}\text{ Br}^-\text{ ions}}$$

$$\text{Moles K}^+ = (3.78 \times 10^{21}\text{ K}^+) \left( \frac{1\text{ mol K}^+}{6.022 \times 10^{23}\text{ K}^+} \right) = 6.27698 \times 10^{-3} = \mathbf{6.28 \times 10^{-3}\text{ mol K}^+}$$

$$\text{Moles Br}^- = (3.78 \times 10^{21}\text{ Br}^-) \left( \frac{1\text{ mol Br}^-}{6.022 \times 10^{23}\text{ Br}^-} \right) = 6.27698 \times 10^{-3} = \mathbf{6.28 \times 10^{-3}\text{ mol Br}^-}$$

- 4.20 Plan: The acids in this problem are all strong acids, so you can assume that all acid molecules dissociate completely to yield  $\text{H}^+$  ions and associated anions. One mole of  $\text{HClO}_4$ ,  $\text{HNO}_3$  and  $\text{HCl}$  each produce one mole of  $\text{H}^+$  upon dissociation, so moles  $\text{H}^+$  = moles acid. Molarity is expressed as moles/L instead of as  $M$ .

Solution:

a) Moles  $\text{H}^+$  = mol  $\text{HClO}_4$  =  $(1.40\text{ L})(0.25\text{ mol/L}) = \mathbf{0.35\text{ mol H}^+}$

b) Moles  $\text{H}^+$  = mol  $\text{HNO}_3$  =  $(1.8\text{ mL})(10^{-3}\text{ L/1 mL})(0.72\text{ mol/L}) = 1.296 \times 10^{-3} = \mathbf{1.3 \times 10^{-3}\text{ mol H}^+}$

c) Moles  $\text{H}^+$  = mol  $\text{HCl}$  =  $(7.6\text{ L})(0.056\text{ mol/L}) = 0.4256 = \mathbf{0.43\text{ mol H}^+}$

- 4.24 Ions in solution that do not participate in the reaction do not appear in a net ionic equation. These spectator ions remain as dissolved ions throughout the reaction. These ions are only present to balance charge.



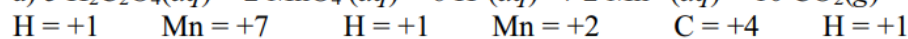
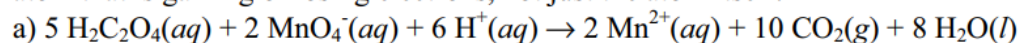
- 4.29 Plan: Check to see if any of the ion pairs are not soluble according to the solubility rules.
- a) Molecular:  $\text{Hg}_2(\text{NO}_3)_2(aq) + 2 \text{KI}(aq) \rightarrow \text{Hg}_2\text{I}_2(s) + 2 \text{KNO}_3(aq)$   
 Total ionic:  $\text{Hg}_2^{2+}(aq) + 2 \text{NO}_3^-(aq) + 2 \text{K}^+(aq) + 2 \text{I}^-(aq) \rightarrow \text{Hg}_2\text{I}_2(s) + 2 \text{K}^+(aq) + 2 \text{NO}_3^-(aq)$   
 Net ionic:  $\text{Hg}_2^{2+}(aq) + 2 \text{I}^-(aq) \rightarrow \text{Hg}_2\text{I}_2(s)$   
 Spectator ions are  $\text{K}^+$  and  $\text{NO}_3^-$ .
- b) Molecular:  $\text{FeSO}_4(aq) + \text{Ba}(\text{OH})_2(aq) \rightarrow \text{Fe}(\text{OH})_2(s) + \text{BaSO}_4(s)$   
 Total ionic:  $\text{Fe}^{2+}(aq) + \text{SO}_4^{2-}(aq) + \text{Ba}^{2+}(aq) + 2 \text{OH}^-(aq) \rightarrow \text{Fe}(\text{OH})_2(s) + \text{BaSO}_4(s)$   
 Net ionic: This is the same as the total ionic equation, because there are no spectator ions.
- 4.31 Plan: A precipitate forms if reactant ions can form combinations that are insoluble, as determined by the solubility rules in Table 4.1. Create cation-anion combinations other than the original reactants and determine if they are insoluble.
- Solution:
- a) No precipitate will form. The ions  $\text{Na}^+$  and  $\text{SO}_4^{2-}$  will not form an insoluble salt according to solubility rule #1: *All common compounds of Group 1A ions are soluble*. The ions  $\text{Cu}^{2+}$  and  $\text{NO}_3^-$  will not form an insoluble salt according to the solubility rule #2: *All common nitrates are soluble*.
- b) A precipitate will form because silver ions,  $\text{Ag}^+$ , and iodide ions,  $\text{I}^-$ , will combine to form a solid salt, silver iodide,  $\text{AgI}$ . The ammonium and nitrate ions do not form a precipitate.
- Molecular:  $\text{NH}_4\text{I}(aq) + \text{AgNO}_3(aq) \rightarrow \text{AgI}(s) + \text{NH}_4\text{NO}_3(aq)$   
 Total ionic:  $\text{NH}_4^+(aq) + \text{I}^-(aq) + \text{Ag}^+(aq) + \text{NO}_3^-(aq) \rightarrow \text{AgI}(s) + \text{NH}_4^+(aq) + \text{NO}_3^-(aq)$   
 Net ionic:  $\text{Ag}^+(aq) + \text{I}^-(aq) \rightarrow \text{AgI}(s)$
- 4.33 Plan: A precipitate forms if reactant ions can form combinations that are insoluble, as determined by the solubility rules in Table 4.1. Create cation-anion combinations other than the original reactants and determine if they are insoluble.
- Solution:
- a) New cation-anion combinations are potassium nitrate and iron(II) chloride. The rules state that all common nitrates and chlorides (with some exceptions) are soluble, so no precipitate forms.
- b) New cation-anion combinations are ammonium chloride and barium sulfate. Again, the rules state that most chlorides are soluble; however, another rule states that sulfate compounds containing barium are insoluble. Barium sulfate is a precipitate and its formula is  $\text{BaSO}_4$ .
- Molecular:  $(\text{NH}_4)_2\text{SO}_4(aq) + \text{BaCl}_2(aq) \rightarrow \text{BaSO}_4(s) + 2 \text{NH}_4\text{Cl}(aq)$   
 Total ionic:  $2 \text{NH}_4^+(aq) + \text{SO}_4^{2-}(aq) + \text{Ba}^{2+}(aq) + 2 \text{Cl}^-(aq) \rightarrow \text{BaSO}_4(s) + 2 \text{NH}_4^+(aq) + 2 \text{Cl}^-(aq)$   
 Net ionic:  $\text{Ba}^{2+}(aq) + \text{SO}_4^{2-}(aq) \rightarrow \text{BaSO}_4(s)$
- 4.35 Plan: Write a balanced equation for the chemical reaction described in the problem. By applying the solubility rules to the two possible products ( $\text{NaNO}_3$  and  $\text{PbI}_2$ ), determine that  $\text{PbI}_2$  is the precipitate. By using molar relationships, determine how many moles of  $\text{Pb}(\text{NO}_3)_2$  (and thus  $\text{Pb}^{2+}$  ion) are required to produce 0.628 g of  $\text{PbI}_2$ .
- Solution:
- The reaction is:  $\text{Pb}(\text{NO}_3)_2(aq) + 2 \text{NaI}(aq) \rightarrow \text{PbI}_2(s) + 2 \text{NaNO}_3(aq)$ .
- $$M \text{ Pb}^{2+} = (0.628 \text{ g PbI}_2) \left( \frac{1 \text{ mol PbI}_2}{461.0 \text{ g PbI}_2} \right) \left( \frac{1 \text{ mol Pb}^{2+}}{1 \text{ mol PbI}_2} \right) \left( \frac{1}{35.0 \text{ mL}} \right) \left( \frac{1 \text{ mL}}{10^{-3} \text{ L}} \right)$$
- $$= 0.0389216 = \mathbf{0.0389 \text{ M Pb}^{2+}}$$

- 4.37 Plan: The balanced equation for this reaction is  $\text{AgNO}_3(aq) + \text{Cl}^-(aq) \rightarrow \text{AgCl}(s) + \text{NO}_3^-(aq)$ . First, determine the moles of  $\text{Cl}^-$  present in the 25.00 mL sample. Second, convert moles of  $\text{Cl}^-$  into grams, and convert the sample volume into grams using the given density. The mass percent of  $\text{Cl}^-$  is found by dividing one result by the other.
- Solution:
- $$(43.63 \text{ mL}) \left( \frac{10^{-3} \text{ L}}{1 \text{ mL}} \right) \left( \frac{0.3020 \text{ mol AgNO}_3}{\text{L}} \right) \left( \frac{1 \text{ mol Cl}^-}{1 \text{ mol AgNO}_3} \right) \left( \frac{35.45 \text{ g Cl}}{1 \text{ mol Cl}^-} \right) = 0.467098 \text{ g Cl (unrounded)}$$
- $$\text{Mass of sample} = (25.00 \text{ mL}) \left( \frac{1.04 \text{ g}}{\text{mL}} \right) = 26.0 \text{ g sample}$$
- $$\text{Mass\% Cl} = \frac{\text{Mass Cl}}{\text{Mass Sample}} \times 100\% = \frac{0.467098 \text{ g Cl}}{26.0 \text{ g Sample}} \times 100\% = 1.79653 = \mathbf{1.80\% \text{ Cl}}$$
- 4.45 a) Molecular equation:  $\text{KOH}(aq) + \text{HI}(aq) \rightarrow \text{KI}(aq) + \text{H}_2\text{O}(l)$   
 Total ionic equation:  $\text{K}^+(aq) + \text{OH}^-(aq) + \text{H}^+(aq) + \text{I}^-(aq) \rightarrow \text{K}^+(aq) + \text{I}^-(aq) + \text{H}_2\text{O}(l)$   
 Net ionic equation:  $\text{OH}^-(aq) + \text{H}^+(aq) \rightarrow \text{H}_2\text{O}(l)$   
 The spectator ions are  $\text{K}^+(aq)$  and  $\text{I}^-(aq)$   
 b) Molecular equation:  $\text{NH}_3(aq) + \text{HCl}(aq) \rightarrow \text{NH}_4\text{Cl}(aq)$   
 Total ionic equation:  $\text{NH}_3(aq) + \text{H}^+(aq) + \text{Cl}^-(aq) \rightarrow \text{NH}_4^+(aq) + \text{Cl}^-(aq)$   
 $\text{NH}_3$  is a weak base and is written in the molecular form.  $\text{HCl}$  is a strong acid and is written in the dissociated form.  $\text{NH}_4\text{Cl}$  is a soluble compound, because all ammonium compounds are soluble.  
 Net ionic equation:  $\text{NH}_3(aq) + \text{H}^+(aq) \rightarrow \text{NH}_4^+(aq)$   
 $\text{Cl}^-$  is the only spectator ion.
- 4.49 The reaction is:  $\text{KOH}(aq) + \text{CH}_3\text{COOH}(aq) \rightarrow \text{KCH}_3\text{COO}(aq) + \text{H}_2\text{O}(l)$
- $$M = \left( \frac{0.1180 \text{ mol KOH}}{\text{L}} \right) \left( \frac{10^{-3} \text{ L}}{1 \text{ mL}} \right) (25.98 \text{ mL}) \left( \frac{1 \text{ mol CH}_3\text{COOH}}{1 \text{ mol KOH}} \right) \left( \frac{1}{52.50 \text{ mL}} \right) \left( \frac{1 \text{ mL}}{10^{-3} \text{ L}} \right)$$
- $$= 0.05839314 = \mathbf{0.05839 \text{ M CH}_3\text{COOH}}$$
- 4.58 a) The S in  $\text{SO}_4^{2-}$  (i.e.,  $\text{H}_2\text{SO}_4$ ) has O.N. = +6, and in  $\text{SO}_2$ , O.N. (s) = +4, so the S has been reduced (and the I<sup>-</sup> oxidized), so the  $\text{H}_2\text{SO}_4$  is an oxidizing agent.  
 b) The oxidation numbers remain constant throughout;  $\text{H}_2\text{SO}_4$  transfers a proton to  $\text{F}^-$  to produce  $\text{HF}$ , so it acts as an acid.
- 4.60 Refer to the rules in Table 4.3. Remember that oxidation number (O.N.) is not the same thing as ionic charge. The O.N. is the charge an atom would have **if** electrons were transferred completely. The compounds in this problem contain covalent bonds, or a combination of ionic and covalent bonds ( $\text{Na}_2\text{C}_2\text{O}_4$ ). Covalent bonds do not involve the complete transfer of electrons.
- a)  $\text{CF}_2\text{Cl}_2$ . The rules dictate that F and Cl each have an O.N. of -1; two F and two Cl yield a sum of -4, so C must have a +4 O.N. **C = +4**
- b)  $\text{Na}_2\text{C}_2\text{O}_4$ . The rule dictates that Na has a +1 O.N.; rule 5 dictates that O has a -2 O.N.; two Na and four O yield a sum of -6 [(+2) + (-8)]. Therefore, the total of the O.N.'s on the two C atoms is +6 and each C is +3. **C = +3**
- c)  $\text{HCO}_3^-$ . H is combined with nonmetals and has an O.N. of +1; O has an O.N. of -2. To have an overall oxidation state equal to -1, C must be +4 because (+1) + (+4) + (-6) = -1. **C = +4**
- d)  $\text{C}_2\text{H}_6$ . Each H has an O.N. of +1; six H gives +6. The sum of O.N.'s for the two C atoms must be -6, so each C is -3. **C = -3**
- 4.62 a)  $\text{NH}_2\text{OH}$ : (O.N. for N) + 3(+1 for H) + 1(-2 for O) = 0      O.N. for N = **-1**  
 b)  $\text{N}_2\text{H}_4$ : 2(O.N. for N) + 4(+1 for H) = 0      O.N. for N = **-2**  
 c)  $\text{NH}_4^+$ : (O.N. for N) + 4(+1 for H) = +1      O.N. for N = **-3**  
 d)  $\text{HNO}_2$ : (O.N. for N) + 1(+1 for H) + 2(-2 for O) = 0      O.N. for N = **+3**

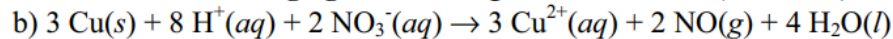
- 4.64 a) AsH<sub>3</sub>. H is combined with a nonmetal, so its O.N. is +1 (Rule 3). The O.N. for As is -3.  
 b) H<sub>3</sub>AsO<sub>4</sub>. The H's in this formula are acidic hydrogens. When an acid dissociates in water, it forms H<sub>3</sub>O<sup>+</sup> or H<sup>+</sup> ion, therefore the O.N. of H in this compound is +1, or +3 for 3 H's. Oxygen's O.N. is -2, with total O.N. of -8 (4 times -2) so As needs to have an O.N. of +5.  
 c) AsCl<sub>3</sub>. Cl has an O.N. of -1, total of -3, so As must have an O.N. of +3.  
 a) **As = -3**      b) **As = +5**      c) **As = +3**

- 4.66 a) MnO<sub>4</sub><sup>2-</sup>: (O.N. for Mn) + 4(-2 for O) = -2      O.N. for Mn = **+6**  
 b) Mn<sub>2</sub>O<sub>3</sub>: {2(O.N. for Mn)} + 3(-2 for O) = 0      O.N. for Mn = **+3**  
 c) KMnO<sub>4</sub>: 1(+1 for K) + (O.N. for Mn) + 4(-2 for O) = 0      O.N. for Mn = **+7**

- 4.68 Oxidizing agent: substance that accepts the electrons released by the substance that is oxidized.  
 Reducing agent: substance that provides the electrons accepted by the substance that is reduced.  
 First, assign oxidation numbers to all atoms. Second, recognize that the agent is the compound that contains the atom that is gaining or losing electrons, not just the atom itself.



Hydrogen and oxygen do not change oxidation state. The Mn changes from +7 to +2 (reduction). Therefore, **MnO<sub>4</sub><sup>-</sup> is the oxidizing agent**. C changes from +3 to +4 (oxidation), so **H<sub>2</sub>C<sub>2</sub>O<sub>4</sub> is the reducing agent**.



Cu changes from 0 to +2 (is oxidized) and **Cu is the reducing agent**. N changes from +5 (in NO<sub>3</sub><sup>-</sup>) to +2 (in NO) and is reduced, so **NO<sub>3</sub><sup>-</sup> is the oxidizing agent**.

- 4.70 a) **Oxidizing agent is NO<sub>3</sub><sup>-</sup>** because nitrogen changes from +5 O.N. in NO<sub>3</sub><sup>-</sup> to +4 O.N. in NO<sub>2</sub>. **Reducing agent is Sn** because its O.N. changes from 0 as the element to +4 in SnCl<sub>6</sub><sup>2-</sup>.  
 b) **Oxidizing agent is MnO<sub>4</sub><sup>-</sup>** because manganese changes from +7 O.N. in MnO<sub>4</sub><sup>-</sup> to +2 O.N. in Mn<sup>2+</sup>. **Reducing agent is Cl<sup>-</sup>** because its O.N. changes from -1 in Cl<sup>-</sup> to 0 as the element to Cl<sub>2</sub>.

- 4.76 Plan: Determine the moles of permanganate ion present. The moles of permanganate ion along with the balanced chemical equation give the moles of hydrogen peroxide. The molar mass of hydrogen peroxide is required to finish the problem. The unrounded numbers from each step are carried to the next step to minimize rounding errors.

Solution:

a) Moles MnO<sub>4</sub><sup>-</sup> =  $\left( \frac{0.105 \text{ mol MnO}_4^-}{\text{L}} \right) \left( \frac{10^{-3} \text{ L}}{1 \text{ mL}} \right) (43.2 \text{ mL}) = 4.536 \times 10^{-3} = \mathbf{4.54 \times 10^{-3} \text{ mol MnO}_4^-}$

b) Moles H<sub>2</sub>O<sub>2</sub> =  $\left( 4.536 \times 10^{-3} \text{ MnO}_4^- \right) \left( \frac{5 \text{ mol H}_2\text{O}_2}{2 \text{ mol MnO}_4^-} \right) = 0.01134 = \mathbf{0.0113 \text{ mol H}_2\text{O}_2}$

c) Mass H<sub>2</sub>O<sub>2</sub> =  $\left( 0.01134 \text{ mol H}_2\text{O}_2 \right) \left( \frac{34.02 \text{ g H}_2\text{O}_2}{1 \text{ mol H}_2\text{O}_2} \right) = 0.3857868 = \mathbf{0.386 \text{ g H}_2\text{O}_2}$

d) Mass percent H<sub>2</sub>O<sub>2</sub> =  $\left( \frac{0.3857868 \text{ g H}_2\text{O}_2}{13.8 \text{ g Sample}} \right) \times 100\% = 2.79556 = \mathbf{2.80\% \text{ H}_2\text{O}_2}$

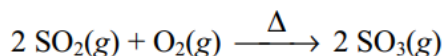
e) **H<sub>2</sub>O<sub>2</sub>** (The hydrogen peroxide reduces the manganese from +7 to +2.)



- 4.83 a)  $\text{Ca}(s) + 2 \text{H}_2\text{O}(l) \rightarrow \text{Ca}(\text{OH})_2(aq) + \text{H}_2(g)$   
**Displacement:** one Ca atom displaces 2 H atoms.  
 b)  $2 \text{NaNO}_3(s) \rightarrow 2 \text{NaNO}_2(s) + \text{O}_2(g)$   
**Decomposition:** one reactant breaks into two products.  
 c)  $\text{C}_2\text{H}_2(g) + 2 \text{H}_2(g) \rightarrow \text{C}_2\text{H}_6(g)$   
**Combination:** reactants combine to form one product.

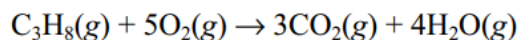
- 4.87 a) The combination between a metal and a nonmetal gives a binary ionic compound.  
 $\text{Ca}(s) + \text{Br}_2(l) \rightarrow \text{CaBr}_2(s)$   
 b) Many metal oxides release oxygen gas upon thermal decomposition.  
 $2 \text{Ag}_2\text{O}(s) \xrightarrow{\Delta} 4 \text{Ag}(s) + \text{O}_2(g)$   
 c) This is a single displacement reaction. Mn is a more reactive metal and displaces  $\text{Cu}^{2+}$  from solution.  
 $\text{Mn}(s) + \text{Cu}(\text{NO}_3)_2(aq) \rightarrow \text{Mn}(\text{NO}_3)_2(aq) + \text{Cu}(s)$

- 4.91 a) Cs, a metal, and  $\text{I}_2$ , a nonmetal, react to form the binary ionic compound, CsI.  
 $2 \text{Cs}(s) + \text{I}_2(s) \rightarrow 2 \text{CsI}(s)$   
 b) Al is a stronger reducing agent than Mn and is able to displace Mn from solution, i.e., cause the reduction from  $\text{Mn}^{2+}(aq)$  to  $\text{Mn}^0(s)$ .  
 $2 \text{Al}(s) + 3 \text{MnSO}_4(aq) \rightarrow \text{Al}_2(\text{SO}_4)_3(aq) + 3 \text{Mn}(s)$   
 c) Sulfur dioxide,  $\text{SO}_2$ , is a nonmetal oxide that reacts with oxygen,  $\text{O}_2$ , to form the higher oxide,  $\text{SO}_3$ .

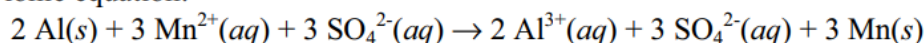


It is not clear from the problem, but energy must be added to force this reaction to proceed.

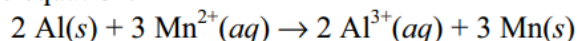
d) Propane is a three-carbon hydrocarbon with the formula  $\text{C}_3\text{H}_8$ . It burns in the presence of oxygen,  $\text{O}_2$ , to form carbon dioxide gas and water vapor. Although this is a redox reaction that could be balanced using the oxidation number method, it is easier to balance by considering only atoms on either side of the equation. First balance carbon and hydrogen (because they only appear in one species on each side of the equation), and then balance oxygen.



e) Total ionic equation:



Net ionic equation:



Note that the molar coefficients are not simplified because the number of electrons lost ( $6 e^-$ ) must equal the electrons gained ( $6 e^-$ ).

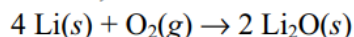
- 4.93 The balanced chemical equation is  $2 \text{HgO}(s) \xrightarrow{\Delta} 2 \text{Hg}(l) + \text{O}_2(g)$

$$\text{Mass O}_2 = (4.27 \text{ kg HgO}) \left( \frac{10^3 \text{ g}}{1 \text{ kg}} \right) \left( \frac{1 \text{ mol HgO}}{216.6 \text{ g HgO}} \right) \left( \frac{1 \text{ mol O}_2}{2 \text{ mol HgO}} \right) \left( \frac{32.00 \text{ g O}_2}{1 \text{ mol O}_2} \right) = 315.420 = \mathbf{315 \text{ g O}_2}$$

The other product is **mercury**.

$$\begin{aligned} \text{Mass Hg} &= (4.27 \text{ kg HgO}) \left( \frac{10^3 \text{ g}}{1 \text{ kg}} \right) \left( \frac{1 \text{ mol HgO}}{216.6 \text{ g HgO}} \right) \left( \frac{2 \text{ mol Hg}}{2 \text{ mol HgO}} \right) \left( \frac{200.6 \text{ g Hg}}{1 \text{ mol Hg}} \right) \left( \frac{1 \text{ kg}}{10^3 \text{ g}} \right) \\ &= 3.95458 = \mathbf{3.95 \text{ kg Hg}} \end{aligned}$$

- 4.95 To determine the reactant in excess, write the balanced equation (metal + O<sub>2</sub> → metal oxide), convert reactant masses to moles, and use molar ratios to see which reactant makes the smaller (“limiting”) amount of product.



$$\text{a) Moles Li}_2\text{O if Li limiting} = (1.62 \text{ g Li}) \left( \frac{1 \text{ mol Li}}{6.941 \text{ g Li}} \right) \left( \frac{2 \text{ mol Li}_2\text{O}}{4 \text{ mol Li}} \right) = 0.1166979 \text{ mol Li}_2\text{O (unrounded)}$$

$$\text{Moles Li}_2\text{O if O}_2 \text{ limiting} = (6.00 \text{ g O}_2) \left( \frac{1 \text{ mol O}_2}{32.00 \text{ g O}_2} \right) \left( \frac{2 \text{ mol Li}_2\text{O}}{1 \text{ mol O}_2} \right) = 0.375 \text{ mol Li}_2\text{O (unrounded)}$$

Li is the limiting reactant; **O<sub>2</sub> is in excess.**

$$\text{b) } 0.1166979 = \mathbf{0.117 \text{ mol Li}_2\text{O}}$$

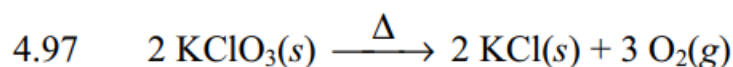
c) Li is limiting, thus there will be none remaining (**0 g Li**).

$$\text{Grams Li}_2\text{O} = (1.62 \text{ g Li}) \left( \frac{1 \text{ mol Li}}{6.941 \text{ g Li}} \right) \left( \frac{2 \text{ mol Li}_2\text{O}}{4 \text{ mol Li}} \right) \left( \frac{29.88 \text{ g Li}_2\text{O}}{1 \text{ mol Li}_2\text{O}} \right) = 3.4869 = \mathbf{3.49 \text{ g Li}_2\text{O}}$$

$$\text{Grams O}_2 \text{ used} = (1.62 \text{ g Li}) \left( \frac{1 \text{ mol Li}}{6.941 \text{ g Li}} \right) \left( \frac{1 \text{ mol O}_2}{4 \text{ mol Li}} \right) \left( \frac{32.00 \text{ g O}_2}{1 \text{ mol O}_2} \right) = 1.867166 \text{ g O}_2 \text{ (unrounded)}$$

The beginning part of these calculations is repeated from the limiting reactant calculation to emphasize that the second part of the problem is simply an extension of the first part. There is no need to repeat the entire calculation as only the final step(s) times the answer of the first part will give the final answer to this part.

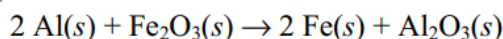
$$\text{Remaining O}_2 = 6.00 \text{ g O}_2 - 1.867166 \text{ g O}_2 = 4.13283 = \mathbf{4.13 \text{ g O}_2}$$



$$\begin{aligned} \text{Mass KClO}_3 &= ((0.900 - 0.700) \text{ g O}_2) \left( \frac{1 \text{ mol O}_2}{32.00 \text{ g O}_2} \right) \left( \frac{2 \text{ mol KClO}_3}{3 \text{ mol O}_2} \right) \left( \frac{122.55 \text{ g KClO}_3}{1 \text{ mol KClO}_3} \right) \\ &= 0.510625 \text{ g KClO}_3 \text{ (unrounded)} \end{aligned}$$

$$\text{Mass\% KClO}_3 = \left( \frac{0.510625 \text{ g KClO}_3}{0.900 \text{ g Sample}} \right) \times 100\% = 56.7361 = \mathbf{56.7\% \text{ KClO}_3}$$

- 4.99 To find the mass of Fe, write a balanced equation for the reaction, determine whether Al or Fe<sub>2</sub>O<sub>3</sub> is the limiting reactant, and convert to mass.



When the masses of both reactants are given, you must determine which reactant is limiting.

$$\text{Mole Fe (from Al)} = (1.00 \text{ kg Al}) \left( \frac{10^3 \text{ g}}{1 \text{ kg}} \right) \left( \frac{1 \text{ mol Al}}{26.98 \text{ g Al}} \right) \left( \frac{2 \text{ mol Fe}}{2 \text{ mol Al}} \right) = 37.064 \text{ mol Fe (unrounded)}$$

$$\text{Mole Fe (from Fe}_2\text{O}_3) = (2.00 \text{ mol Fe}_2\text{O}_3) \left( \frac{2 \text{ mol Fe}}{1 \text{ mol Fe}_2\text{O}_3} \right) = 4.00 \text{ mol Fe}$$

Fe<sub>2</sub>O<sub>3</sub> is limiting, so 4.00 moles of Fe forms.

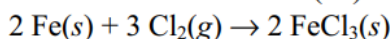
$$\text{Mass} = (4.00 \text{ mol Fe}) \left( \frac{55.85 \text{ g Fe}}{1 \text{ mol Fe}} \right) = 223.4 = 223 \text{ g Fe}$$

Though not required by the problem, this could be converted to **0.223 kg**.

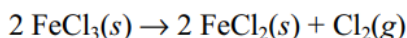


- 4.100 In ionic compounds, iron has two common oxidation states, +2 and +3. First, write the balanced equations for the formation and decomposition of compound A. Then, determine which reactant is limiting and from the amount of the limiting reactant calculate how much compound B will form.

Compound A is iron chloride with iron in the higher, +3, oxidation state. Thus the formula for compound A is  $\text{FeCl}_3$  and the correct name is iron(III) chloride. The balanced equation for formation of  $\text{FeCl}_3$  is:



The  $\text{FeCl}_3$  decomposes to  $\text{FeCl}_2$  with iron in the +2 oxidation state. The balanced equation for the decomposition of  $\text{FeCl}_3$  is:



To find out whether 50.6 g Fe or 83.8 g  $\text{Cl}_2$  limits the amount of product, calculate the number of moles of iron(III) chloride that could form based on each reactant.

$$\text{Mole FeCl}_3 (\text{from Fe}) = (50.6 \text{ g Fe}) \left( \frac{1 \text{ mol Fe}}{55.85 \text{ g Fe}} \right) \left( \frac{2 \text{ mol FeCl}_3}{2 \text{ mol Fe}} \right) = 0.905998 \text{ FeCl}_3 (\text{unrounded})$$

$$\text{Mole FeCl}_3 (\text{from Cl}_2) = (83.8 \text{ g Cl}_2) \left( \frac{1 \text{ mol Cl}_2}{70.90 \text{ g Cl}_2} \right) \left( \frac{2 \text{ mol FeCl}_3}{3 \text{ mol Cl}_2} \right) = 0.787964 \text{ FeCl}_3 (\text{unrounded})$$

Since fewer moles of  $\text{FeCl}_3$  are produced from the available amount of chlorine, the chlorine is limiting and we calculate the amount of compound B from the amount of chlorine gas.

$$\begin{aligned} \text{Mass FeCl}_2 &= (83.8 \text{ g Cl}_2) \left( \frac{1 \text{ mol Cl}_2}{70.90 \text{ g Cl}_2} \right) \left( \frac{2 \text{ mol FeCl}_3}{3 \text{ mol Cl}_2} \right) \left( \frac{2 \text{ mol FeCl}_2}{2 \text{ mol FeCl}_3} \right) \left( \frac{126.75 \text{ g FeCl}_2}{1 \text{ mol FeCl}_2} \right) \\ &= 99.874 = \mathbf{99.9 \text{ g FeCl}_2} \end{aligned}$$

- 4.104 On a molecular scale, chemical reactions are dynamic. If NO and  $\text{Br}_2$  are placed in a container, molecules of NO and molecules of  $\text{Br}_2$  will react to form NOBr. Some of the NOBr molecules will decompose and the resulting NO and  $\text{Br}_2$  molecules will recombine with different NO and  $\text{Br}_2$  molecules. In this sense, the reaction is dynamic because the original NO and  $\text{Br}_2$  pairings do not remain permanently attached to each other. Eventually, the rate of the forward reaction (combination of NO and  $\text{Br}_2$ ) will equal the rate of the reverse reaction (decomposition of NOBr) at which point the reaction is said to have reached equilibrium. If you could take “snapshot” pictures of the molecules at equilibrium, you would see a constant number of reactant (NO,  $\text{Br}_2$ ) and product molecules (NOBr), but the pairings would not stay the same.

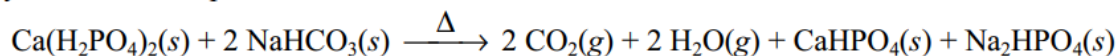
- 4.106 a)  $\text{Fe}(s) + 2 \text{H}^+(aq) \rightarrow \text{Fe}^{2+}(aq) + \text{H}_2(g)$   
 O.N.: 0 +1 +2 0

$$\begin{aligned} \text{b) Fe}^{2+} \text{ ions} &= (737 \text{ g Sauce}) \left( \frac{49 \text{ mg Fe}}{125 \text{ g Sauce}} \right) \left( \frac{10^{-3} \text{ g}}{1 \text{ mg}} \right) \left( \frac{1 \text{ mol Fe}}{55.85 \text{ g Fe}} \right) \left( \frac{1 \text{ mol Fe}^{2+}}{1 \text{ mol Fe}} \right) \left( \frac{6.022 \times 10^{23} \text{ Fe}^{2+} \text{ ions}}{1 \text{ mol Fe}^{2+}} \right) \\ &= 3.11509 \times 10^{21} = \mathbf{3.1 \times 10^{21} \text{ Fe}^{2+} \text{ ions}} \text{ per jar of sauce} \end{aligned}$$

- 4.119 The reaction is:  $2 \text{Na}_2\text{O}_2(s) + 2 \text{CO}_2(g) \rightarrow 2 \text{Na}_2\text{CO}_3(s) + \text{O}_2(g)$ .

$$\begin{aligned} \text{Volume} &= (80.0 \text{ g Na}_2\text{O}_2) \left( \frac{1 \text{ mol Na}_2\text{O}_2}{77.98 \text{ g Na}_2\text{O}_2} \right) \left( \frac{2 \text{ mol CO}_2}{2 \text{ mol Na}_2\text{O}_2} \right) \left( \frac{44.01 \text{ g CO}_2}{1 \text{ mol CO}_2} \right) \left( \frac{\text{L Air}}{0.0720 \text{ g CO}_2} \right) \\ &= 627.08 = \mathbf{627 \text{ L Air}} \end{aligned}$$

- 4.126 Balance the equation to obtain the correct molar ratios. This is not a redox reaction as none of the O.N.'s change. Here is a suggested method for approaching balancing the equation.
- Since  $\text{PO}_4^{2-}$  remains as a unit on both sides of the equation, treat it as a unit when balancing.
  - On first inspection, one can see that Na needs to be balanced by adding a “2” in front of  $\text{NaHCO}_3$ . This then affects the balance of C, so add a “2” in front of  $\text{CO}_2$ .
  - Hydrogen is not balanced, so change the coefficient of water to “2,” as this will have the least impact on the other species.
  - Verify that the other species are balanced.



a) Determine whether  $\text{Ca}(\text{H}_2\text{PO}_4)_2$  or  $\text{NaHCO}_3$  limits the production of  $\text{CO}_2$ . In each case calculate the moles of  $\text{CO}_2$  that might form.

$$\begin{aligned} \text{Mole CO}_2 (\text{NaHCO}_3) &= (1.00 \text{ g}) \left( \frac{31\%}{100\%} \right) \left( \frac{1 \text{ mol NaHCO}_3}{84.01 \text{ g NaHCO}_3} \right) \left( \frac{2 \text{ mol CO}_2}{2 \text{ mol NaHCO}_3} \right) \\ &= 3.690 \times 10^{-3} \text{ mol CO}_2 \text{ (unrounded)} \end{aligned}$$

$$\begin{aligned} \text{Mole CO}_2 (\text{Ca}(\text{H}_2\text{PO}_4)_2) &= (1.00 \text{ g}) \left( \frac{35\%}{100\%} \right) \left( \frac{1 \text{ mol Ca}(\text{H}_2\text{PO}_4)_2}{234.05 \text{ g Ca}(\text{H}_2\text{PO}_4)_2} \right) \left( \frac{2 \text{ mol CO}_2}{1 \text{ mol Ca}(\text{H}_2\text{PO}_4)_2} \right) \\ &= 2.9908 \times 10^{-3} \text{ mol CO}_2 \text{ (unrounded)} \end{aligned}$$

Since  $\text{Ca}(\text{H}_2\text{PO}_4)_2$  is limiting,  **$3.0 \times 10^{-3} \text{ mol CO}_2$**  will be produced.

$$\text{b) Volume CO}_2 = (2.9908 \times 10^{-3} \text{ mol CO}_2) (37.0 \text{ L/mol CO}_2) = 0.1106596 = \mathbf{0.11 \text{ L CO}_2}$$